Neelkumar Ahir

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Summary

Robotics Engineer specializing in motion planning and controls, robot learning for autonomous systems, with expertise in C++, Python, ROS2, NMPC, and trajectory optimization in translating complex problems into efficient solutions for real-world deployment.

Education

California State University, Sacramento, CA **Aug 2023 – Dec 2025**
Master of Science in Mechanical Engineering *GPA: 3.8/4*

Awards:

- **Best Commercial Potential Award** - Autonomous Robot Build Day at AGI HOUSE
- **Special Project Award** - Physical AI Hackathon 2026
- **Provost's Award (1st Place)** – Sacramento State Spring Symposium, 2025
- **Honorable Mention (3rd Place)** – The 39th Annual CSU Student Research Competition, hosted by Cal Poly Humboldt, *2025*

Gujarat Technological University, Gujarat, India **July 2019 – May 2023**
Bachelor of Engineering in Mechanical Engineering *GPA: 3.1/4*

Experience

Robotics Engineer intern **Feb 2026 – present**
Open Droid *San Francisco, CA*

- **Autonomous Navigation:** Developed and fine-tuned indoor navigation stacks for the R2D3 robot platform using ROS2, focusing on SLAM, costmap inflation, and precise path planning in dynamic environments.
- **LLM Orchestration:** Integrated Gemini API to serve as the robot's "reasoning engine," enabling the agent to interpret complex environmental context and autonomously determine navigation goals.
- **System Integration:** Engineered a communication bridge between high-level Generative AI orchestration and low-level ROS2 controllers, translating natural language reasoning into actionable spatial coordinates.
- **Optimization:** Improved R2D3's decision-making latency by optimizing API call structures and localizing reasoning processes for real-time indoor deployment.

Robotics Software Engineer – Planning and Control **Dec 2024 – Dec 2025**
Competitive Robotics Club – Firefighter Project *California State University, Sacramento*

- **Engineered a multi-tier navigation stack in C++,** implementing the A search algorithm* for global path planning and integrating it with local PID controllers for precise obstacle avoidance and trajectory execution.
- **Optimized path-finding efficiency** by implementing custom heuristic functions within the A* framework, reducing computational overhead and ensuring real-time re-planning capabilities in dynamic environments.
- **Developed a robust state estimation** pipeline by fusing dual IMU data and odometry using a sensor fusion filter, significantly improving localization accuracy during high-acceleration maneuvers.
- **Designed and implemented discrete-time PID** controllers for motor velocity and heading control, tuning gains to achieve minimal overshoot and a settling time.
- **Maintained a professional CI/CD** workflow using **Git/GitHub** for collaborative development; performed rigorous unit testing and Gazebo-based regression tests to validate behavior logic before hardware deployment.

Graduate Teaching Assistant **Aug 2024 – Jan 2025**
California State University, Sacramento *Sacramento, California*

- Assisting/Grading for Advanced Mathematics (ME201, ME202).

Robotics Software Engineer – Controls Intern **Jan 2023 – May 2023**
Larsen and Toubro Defense IC (L&T) *Surat, India*

- Developed high-performance control laws by designing and **tuning multi-axis PID** and **Feedforward controllers** to emulate specific electric drivetrain responses for heavy-duty robotic actuators.
- Modeled complex plant dynamics in **MATLAB/Simulink** to analyze torque-dependent behaviors, ensuring stable actuator performance across varying load conditions and duty cycles.
- **Characterized system response** by performing frequency-domain analysis to identify resonance modes, utilizing this data to optimize controller gains for high-speed precision movement.
- **Bridged simulation and hardware** by validating joint-level dynamics against real-world telemetry, achieving a high degree of correlation between Simulink predictive models and physical hardware output.
- **Engineered a Python-based automation tool** for inspection report processing and tolerance filling, streamlining the quality assurance workflow and reducing manual overhead by 40%.

Technical Skills

Programming Languages: C++, Python, MATLAB/Simulink
Frameworks/Tools: ROS2, Gazebo, Issac Sim/lab, CasADi, Pytorch, Git/GitHub
Simulation & CAD: SolidWorks, CATIA, Ansys, FEA
Hardware Platforms: Jetson AGX Thor, Jetson Orin Nano, Pixhawk(PX4), IMU, Encoders, GPS, Linux, HITL/SITL, SO-101 arm

Projects

ServingU – Physical AI Hackathon 2026 **31st Jan – 01st Feb 2026**

- Utilizing **the LeRobot SO-101** arm, Jetson Orin Nano, mobile base as hardware, and a custom environment for a precise pouring task.
- Divided the whole process of pouring into **3 subtasks**, collected data for each subtask separately, and trained models (ACT, Diffusion) for each subtask on **Brev**.
- **Evaluated and benchmarked the models** and went with the best-performing model (**ACT**).
- At the end, built the **custom script** to automatically run inference for each model sequentially (with an option to manually switch), ensuring the coffee is poured into the serving cup precisely with an over **90%** success rate.

LeSandwitch - Embodied- AI Hackathon by Seeed Studio, NVIDIA, Hugging Face (link) Sep 2025 – Oct 2025

- Built a mobile manipulator with dual **LeRobot SO-101** arms and a modular kitchen setup for sandwich preparation and delivery
- Teleop data collection via **leader–follower** control; expanded datasets with simulation in **IsaacSim/IsaacLab**
- Trained **GR00T-N1.5** and **ACT** policies on cloud GPUs using **Runpod, Vast.ai, Brev, and PhosphoBot**
- Deployed real-time inference on **NVIDIA Jetson AGX Thor** (prototyped on **Jetson Orin Nano Super**).

Thesis: Nonlinear Model Predictive Control of a Quadrotor **Jan 2024 - Present**

- Derived a 6-DOF quadrotor plant model in **MATLAB/Simulink**, incorporating non-linear multi-body dynamics, aerodynamic effects, and physical actuator saturation constraints.
- Developed **a Minimum Snap trajectory generator** for the vertical (Z) axis to ensure smooth, differentiable motion profiles, optimizing for reduced mechanical stress and motor jitter during high-dynamic maneuvers.
- Architected a **Nonlinear Model Predictive Controller (NMPC)** using the **CasADi** optimization framework; engineered a multi-objective cost function to minimize pose and velocity errors while respecting state-space constraints.
- Achieved **86.7% ISE** and **92.6% ITAE** improvement with NMPC over PD, validated in MATLAB.
- Performed experimental flight tests on **DJI Phantom 4 Pro V2.0** across 3 trajectories to validate controller performance.

Autonomous Rover Design **Sep 2024 – Dec 2024**

- **Developed a custom Pure Pursuit controller** in C++ for high-accuracy trajectory tracking, implementing a velocity-indexed look-ahead distance to optimize stability and minimize cross-track error.
- **Architected a ROS 2 navigation stack** using Action Servers for waypoint sequencing and TF2 for real-time coordinate frame transformations between map, odom, and base_link.
- **Engineered an Extended Kalman Filter (EKF)** to fuse asynchronous data from dual IMUs and wheel encoders at 30 Hz, successfully reducing cumulative yaw drift by 55%.
- **Integrated high-level planning nodes** with Pixhawk flight controllers via MAVROS, ensuring low-latency communication and robust GPS-based waypoint navigation in outdoor environments.
- **Validated motion algorithms** through Gazebo simulations and Hardware-in-the-Loop (HIL) testing, refining PID gains to ensure smooth acceleration profiles across varying terrains.

Publication

- Garima Bhandari, **Neelkumar Ahir**, Atefeh Mohammadpour. "Enhancing Drone Trajectory Tracking in Construction via Proportional– Derivative Control Optimization" submitted to the 62nd ASC International Conference, 2026, Accepted.
- **Neelkumar Ahir**, Jesus Betancourt, and Garima Bhandari. "Non-Linear Model Predictive Control of Quadrotor for Trajectory Tracking" under review for IEEE IROS.
- **Neelkumar Ahir,** Jyotindra Narayan and Garima Bhandari, Member "Constraint Aware Nonlinear Model Predictive Control for Quadrotor Trajectory Tracking: Comparison with Optimally Tuned PD Control" under review for IEEE CoDIT.